Meeting brief — Centre-aware population graph construction for cross-institution HNC survival prediction

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1. Problem in one paragraph

Population-graph GNNs predict head-and-neck cancer survival by treating each patient as a node (PET/CT radiomics) and connecting similar patients. The strongest published model (Peng et al., BSPC 2024) builds edges wherever fused PET/CT similarity exceeds a fixed 0.2 threshold. Because scanner and protocol effects make same-centre patients look alike, this rule connects patients mostly within their own centre, so the network learns centre-specific neighbourhoods. Result: C-index 0.788 on held-out TCIA cohorts but **0.674** on centres never present in the graph (HECKTOR 2021). I propose to change only the edge rule — connect each patient to its most similar patients from **every other centre** — and to evaluate with strict leave-one-centre-out, keeping features, GNN and Cox head identical to Peng.

2. How the reference papers lead here

Peng et al. 2024 (BSPC) — the pipeline I reproduce and extend. Documents the 0.2 threshold, the 7-centre TCIA cohort, and the 0.788 → 0.674 drop that motivates the work. Compares four GNN layers but only one graph-construction rule.
Mienye & Viriri 2025 (Information, survey) — names graph construction as data-dependent and unstable (§5.1.1), cross-institution generalisation as unresolved (§5.1.4), and calls for "harmonised graph construction protocols" (§5.2.3). This paper is a direct answer to that call.
Chai et al. 2026 (BSPC) — precedent for k-NN and adaptively chosen neighbourhoods (Vision-GNN Top-K, K-Estimate net) and for releasing code. Justifies k-NN over thresholding.
Zhang et al. 2025 (BSPC) — background: the field has invested in GNN *architecture* for medical imaging while *construction* is under-studied.

3. Method — one change

`centre_balanced_knn(X, centre, k)`: for patient i, select the k most similar patients from each centre $c \neq centre(i)$; edge weight = similarity. Unseen-centre patients at test time are attached to the training graph by the same rule → the model becomes inductive.

Three graphs compared with identical downstream model (GAT + MLP + Cox, as in Peng):

SNF similarity > 0.2 (Peng, reproduced)
Plain k-NN (isolates "k-NN vs threshold")
Centre-balanced k-NN (isolates "centre mixing")

4. Contributions

A centre-balanced k-NN construction rule for population graphs under domain shift.
First leave-one-centre-out benchmark of graph-construction rules for HNC PET/CT survival (Peng used one fixed split).
Reproducible release: code, pinned PyRadiomics config, splits, seeds.

5. Evaluation

Protocol: leave-one-centre-out × 5 seeds; report mean ± std of Harrell's and Uno's C-index, time-dependent AUC, IBS.
Headline figure: in-centre vs held-out-centre C-index per graph type — the **generalisation gap**.
Baselines: clinical Cox, radiomics Cox-PH, Cox-nnet, random survival forest (so reviewers cannot say a tabular model wins).
Statistics: paired bootstrap on C-index differences across folds.

6. Data plan (TCIA, free)

Stage	Collection	Centres	Patients (with PET/CT + outcome, per Peng)
Now	Head-Neck-PET-CT (Vallières et al.)	CHUM, CHUS, HGJ, HMR	296 → 4-fold leave-one-centre-out
Next	Head-Neck-Radiomics-HN1 (MAASTRO), QIN-HEADNECK (Iowa)	+2	+74, +243 → 6 centres
Later	HECKTOR-2022 (on request)	CHUP, CHUV, MDA, USZ …	truly unseen sites for final external test

Access: TCIA NBIA Data Retriever (free registration). Head-Neck-PET-CT ships DICOM PET/CT, RTSTRUCT GTV contours, and a clinical/outcome spreadsheet; the patient ID prefix gives the centre label.

7. Timeline (~10 weeks)

Wk	Milestone
1–2	Download Head-Neck-PET-CT; convert DICOM/RTSTRUCT → NIfTI; PyRadiomics extraction (pinned config, 1 mm isotropic, 64 bins as in Peng); feature table with `centre`, `PFS_time`, `PFS_event`
3–4	Reproduce Peng: feature selection (variance → univariate Cox → Spearman 0.8 → LASSO), SNF-0.2 graph, GAT + Cox; target C-index ≈ 0.77–0.79 on their split

Wk	Milestone
5	`build_graph(mode)` with three modes; leave-one-centre-out loop; inductive attachment of held-out patients
6–7	Full grid: 3 graphs × 4 folds × 5 seeds; sweep k ∈ {2,3,5,8}
8	Tabular baselines; add HN1/QIN if downloaded
9–10	Paper (BSPC format), DOI verification, code release

8. Risks and fallbacks

Reproduction gap (cannot hit ~0.78): report own SNF-0.2 number as baseline; the comparison is still internally fair.
Centre-balanced k-NN does not help: the benchmark and negative result are still publishable — the protocol is new for this task.
Only 4 centres: acceptable for a first paper; HN1/QIN extend to 6, HECKTOR to 10+.
Compute: single GPU sufficient; radiomics extraction is CPU-bound (hours).

9. Questions for the guide

Target venue: BSPC (fits the reference set) vs IEEE Access / Diagnostics (faster review)?
Endpoint: PFS as primary (matches Peng) — also report OS?
Is a 4-centre first study acceptable, with HECKTOR as a planned extension?

10. Immediate next actions (this week)

Register on TCIA, download Head-Neck-PET-CT and its clinical spreadsheet.
Confirm outcome fields available (PFS/OS time + event).
Set up repo: `data/`, `features/`, `graphs/`, `models/`, `eval/`; pin PyRadiomics config.